Advanced Math **Performace Assessment** _____

Multivariable Calculus Applied Projects

Key words: Vector Operations.

Name: _____ **Number:** _____

Project 1: MICROARRAY ANALYSIS OF GENOME EXPRESSION

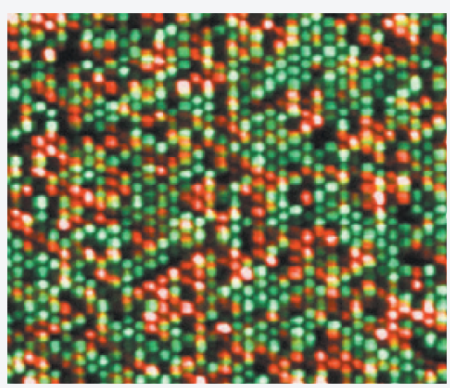
Microarrays provide a technique for quantifying how the expression of different genes in the genome changes in response to changes in conditions. Suppose we are interested in how a drug affects genome expression. Cells are maintained under a reference, or normal, condition as well as in the presence of the drug.

All gene products generated under the reference condition are labeled with a green dye, and all gene products generated in the presence of the drug are labeled with a red dye. The next step is to take all known genes individually and adhere copies of each to different locations on a glass plate. The dyed extracts from cells are then washed over the plate, and each of the dyed gene products binds with its corresponding gene.

The result is a glass plate containing thousands of colored dots, one for each gene. The dots vary in color from green to yellow to red:

- A dot is green if the corresponding gene tends to be expressed less in the presence of the drug than in the reference condition because most of the extract binding to this gene on the plate will have come from reference cells.
- A dot is yellow if the corresponding gene is expressed equally in both cell types,
- A dot is red if the gene tends to be expressed more in the presence of the drug than in the reference condition.

The entire collection of colored dots on the plate is called a microarray. An example is shown in the following figure.



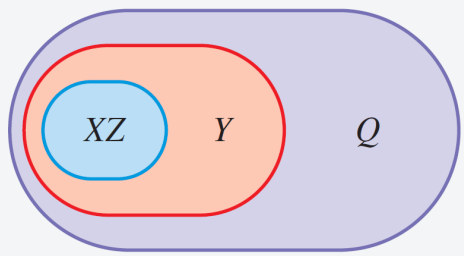
To analyze microarray data the color intensity for each gene is translated into a numerical score in the interval $(-\infty, \infty)$, with negative values indicating green and positive values indicating red. For simplicity, let's suppose there are five genes on the microarray and we have four different potential drugs, giving the following data:

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5
Drug A	1	3	5	6	9
Drug B	2	6	9	12	19
Drug C	-1	6	7	0	3
Drug D	5	2	5	7	12

Suppose our goal is to determine the similarity of drugs A, B, C, and D in terms of the patterns of genome expression that they induce. To do so, we need a measure of similarity of any two profiles.

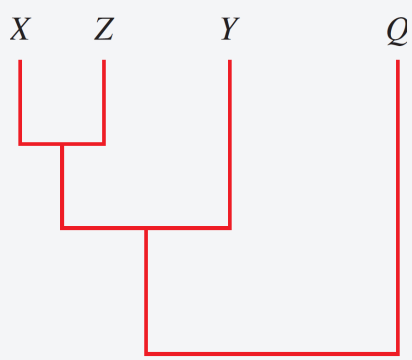
(1) Treating the expression profile for each drug as a point in $\mathbb{R}^5$, calculate the distances between all pairs of profiles.

(2) The next step is to group the drugs hierarchically according to their similarity. To do so, first take the two drugs that are closest and draw a circle around them. Then, take the drug that is next closest to either of these first two and draw a curve that encloses all three. Proceed until all drugs are included, giving a diagram depicting hierarchical clusters of decreasing similarity as shown in the following figure.



Drugs X and Z are most similar, followed by drug Y, and then drug Q.

It is also possible to depict the drug similarities using a dendogram, where the length of the path in the diagram between two drugs reflects the similarity between the drugs



An example of a dendogram corresponding to the clusters in the figure above.

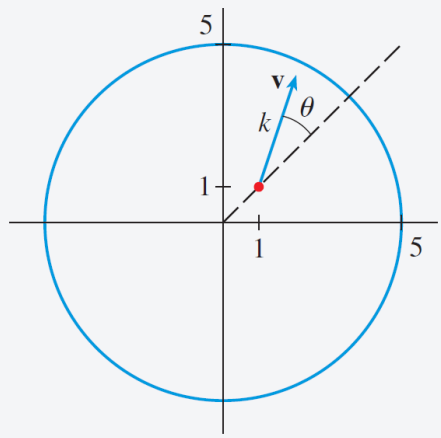
Please construct both.

(3) The other commonly used measure of similarity between profiles involves the dot product. Treating the expression profile for each drug as a vector in V_5, calculate the cosine of the angle between each of the pairs of profiles as a measure of similarity. Repeat the clustering analysis in Problem (2) for this measure.

Problems (1) and (3) provide different measures of the similarity between pairs of profiles, and both are commonly used in microarray analysis. The similarity measure in Problem (1) is sometimes referred to as the Euclidean distance and the similarity measure in Problem (3) is sometimes referred to as the Pearson correlation coefficient.

Project 2: VACCINE ESCAPE

Vaccine escape refers to the loss of effectiveness of a vaccine due to the antigenic evolution of the pathogen. Suppose that a vaccine protects against all pathogens within a circle of radius 5 centered at the origin in two-dimensional antigenic space. Further, suppose that the vector $\boldsymbol{v}$ giving the instantaneous velocity of evolution in antigenic space is always of fixed magnitude k and angle θ from a line connecting the pathogen to the origin, with $0 < \theta < \pi/2$.



(1) Suppose the pathogen is currently at location $(1, 1)$ in antigenic space. Using a vector projection, determine the vector that specifies the velocity at which the pathogen is evolving directly away from the origin.

(2) Suppose the pathogen is currently at location $(1, 1)$ in antigenic space. Using a vector projection, determine the vector that specifies the velocity at which the pathogen is evolving directly away from the origin.

(3) In general, suppose the pathogen is at location (x, y). Using a scalar projection, determine the speed at which the pathogen is evolving directly away from the origin.

(4) Use your result from Problem (3) to write a differential equation that governs the distance of the pathogen from the origin.

(5) Integrate the differential equation from Problem (4) and determine the time at which vaccine escape occurs (that is, the time when the pathogen escapes the vaccine's circle of protection), assuming the distance to the origin at time $t = 0$ is $\sqrt{2}$.